



DPULimiter

The DPULimiter limits the slope and the absolute value of a signal.

1. Inputs:

In Input signal

2. Dynamic parameters:

Min Minimum of output.

Max Maximum of output.

SlopeLimitNeg Limit of negative slope of signal [1/s].

SlopeLimitPos Limit of positive slope of signal [1/s].

3. Outputs:

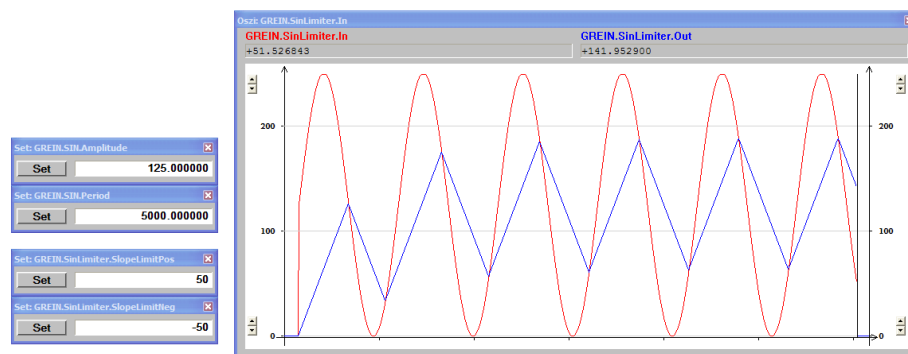
Out limited output signal

4. Functional principle:

If this signal *In* increases or decreases faster than the specified limits, then the output *Out* is just increased (or decreased) about the maximum increase per second (or maximum drop per second).

5. Example:

As input signal *In* a sine oscillation with a period of 5 sec and an amplitude of 125 is used. The boundaries of the limiter are set to +50/s respectively -50/s.

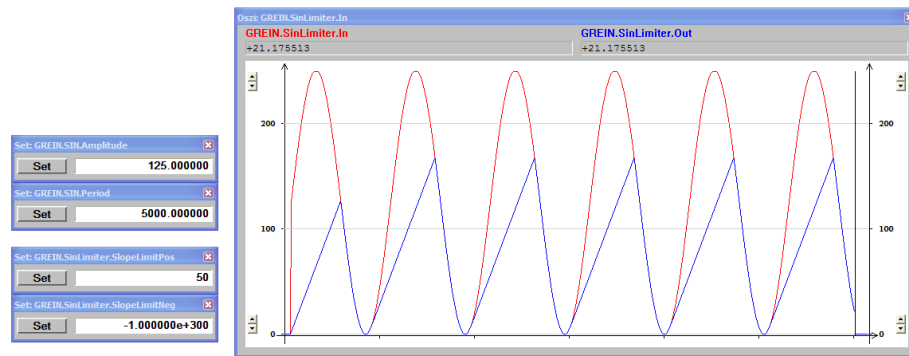


The limited output signal (blue) can't follow the sine and levels out - triangle-shaped - around the mean value.



SILAB DRIVING SIMULATION

version Version 7.1 documentation



In the 2nd case the signal can decrease infinitely fast ($-1e300/s$) and thus follows a part of the sine oscillation.